VSP KONDALARAO M

+91-9866733796 | mantrikr@gmail.com | pavan-mantri-aa842924b | pa1mantri

EDUCATION

Liverpool John Moores University — **2010 – 2012**
Masters in Microelectronics — *Liverpool, UK*

Pragati Engineering College — **2005 – 2009**
B.Tech in Electronics and Communication — *Kakinada, India*

PROJECTS

Tapeout of an Asynchronous CDC FIFO |

- Designed and taped out an 8-bit wide asynchronous FIFO using the sky130 PDK to ensure robust clock domain crossing (CDC), decoupling write and read operations to prevent data loss and metastability between unsynchronized clock domains.
- Optimized silicon area by using a dual-port RAM instead of standard SRAM due to tapeout constraints.

Design of an Application-Specific RISC-V SoC for Clap Switch Control |

- Built a Customizable RISC-V SoC with UART-based instruction bootloader and an on-chip SRAM instruction memory.
- Reduced area from 50K to 15K standard cells by deriving and implementing only the minimal instruction subset required by the application.

Physical Implementation and Integration of a Custom SoC Debug Block |

- Executed the complete RTL-to-GDSII physical design flow for a 32-bit Wishbone slave wrapper using the OpenROAD toolchain, resolving placement density and pin-routing violations.
- Validated the physically implemented netlist using mixed-level Gate-Level Simulation (GLS) within the full Caravel RTL SoC environment to ensure timing-clean system integration.
- **Tech Node & Framework:** SkyWater 130nm PDK, Efabless Caravel SoC, OpenROAD Flow Scripts (ORFS)

Characterization and Integration of a Custom Inverter Cell into the PicoRV32a Core |

- Characterized a custom inverter cell by extracting rise time, fall time, propagation delay, and noise margins using ngspice and Sky130 PDK.
- Integrated the characterized inverter into a custom standard cell library and used it in the full RTL-to-GDS flow of PicoRV32a using Yosys, OpenLane, and Magic.Achieved timing closure using OpenSTA.

Internships

Fossee Winter Internship, IIT Bombay — **Dec 2025 – Feb 2026**

- Designed and simulated Phase-Locked Loop (PLL) using the BiCMOS IHP PDK and eSim tool.
- Integrated the IHP sg13g2 PDK into eSim tool. eSim, an open-source EDA mixed signal tool, is developed under the Ministry of Education's FOSSEE initiative.

Achievements & Hackathons

AI/ML-Powered FPGA Design Hackathon, NIT Jamshedpur — **Dec 2025**
Ranked 25th nationally

- Built a 3-layer hardware neural network accelerator featuring custom activation logic, weight storage, and a configurable datapath supporting runtime model swapping via UART.

TECHNICAL SKILLS

Languages: Verilog, System Verilog, Python, TCL
Tools: Synopsys DC, ICC2, Primetime, Openroad, Yosys, Magic
Technologies/Frameworks: Linux, GitHub,Git

CERTIFICATIONS

- Attended a 5-day workshop on RTL2GDSII flow using Synopsys Tools.
- Semiconductor Fabrication 101 - Purdue University
- VSD Physical Design flow - Udemy.
- VLSI design course - CDAC.
- RISC-V based MYTH processor design - Redwood EDA.